

Is an ontology needed here?

Useful as the shared semantic contract in the current architecture.

What it provides

Explicit product / process / resource links
Reusable assembly relationships
A common structure for PA and RA

What the claim requires

OWL/RDF is a representation choice.
Typed JSON can retain the PPR model.
Accuracy gains need an experiment.

Keep PPR for explicit shared meaning; evaluate its added value separately.

PPR connects Product, Process and Resource

PRODUCT

**What is involved
and required?**

product • feature • state
specification
Assembly / Part extension

medium gear
gear plate assembly

PROCESS

**What activity realizes
the goal?**

process
processExecution
realizes

assembly
recorded assignment

RESOURCE

**Which resources can
be considered?**

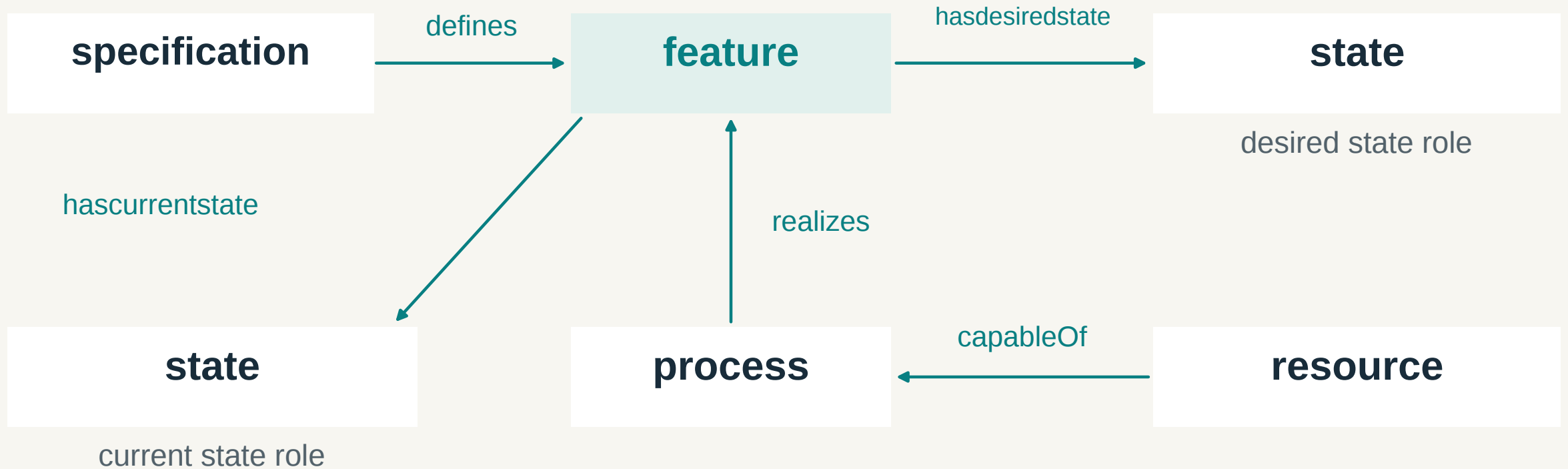
resource
capableOf → process
configured authority

xarm6
ur5e

Broad process capability admits candidates; task-specific planning supports assignment.

capability is declared, but the current Workcell ABox links resource directly to process through capableOf.

The core PPR graph links a requirement to resources



specification → feature ← process ← resource

Schema view: class boxes and property domain → range. Product ownership and assignment are detailed in later slides.

Schema, facts and evidence have different jobs

TBox — supplied schema

Classes, properties, hierarchy
Pairwise cardinality
No task or resource instances

Interaction ABox — this task

specification + target feature
States + assembly relationships
Selected assignment

Workcell ABox — configuration

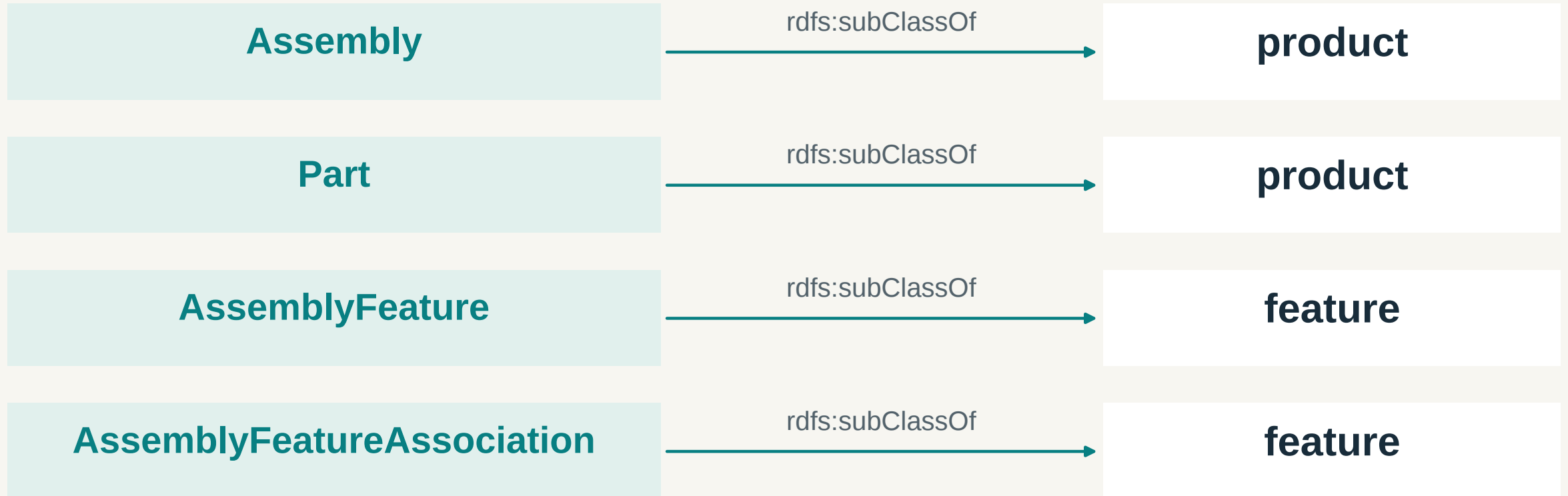
process and resource individuals
resource capableOf process
System-owned capability authority

Linked records — grounding detail

State text, names and value_ref
Typed geometry + uncertainty
Evidence references and hashes

The graph carries explicit relations; linked records preserve their evidence and values.

The assembly extension adds four classes



An assembly relationship can use the existing feature-state vocabulary.

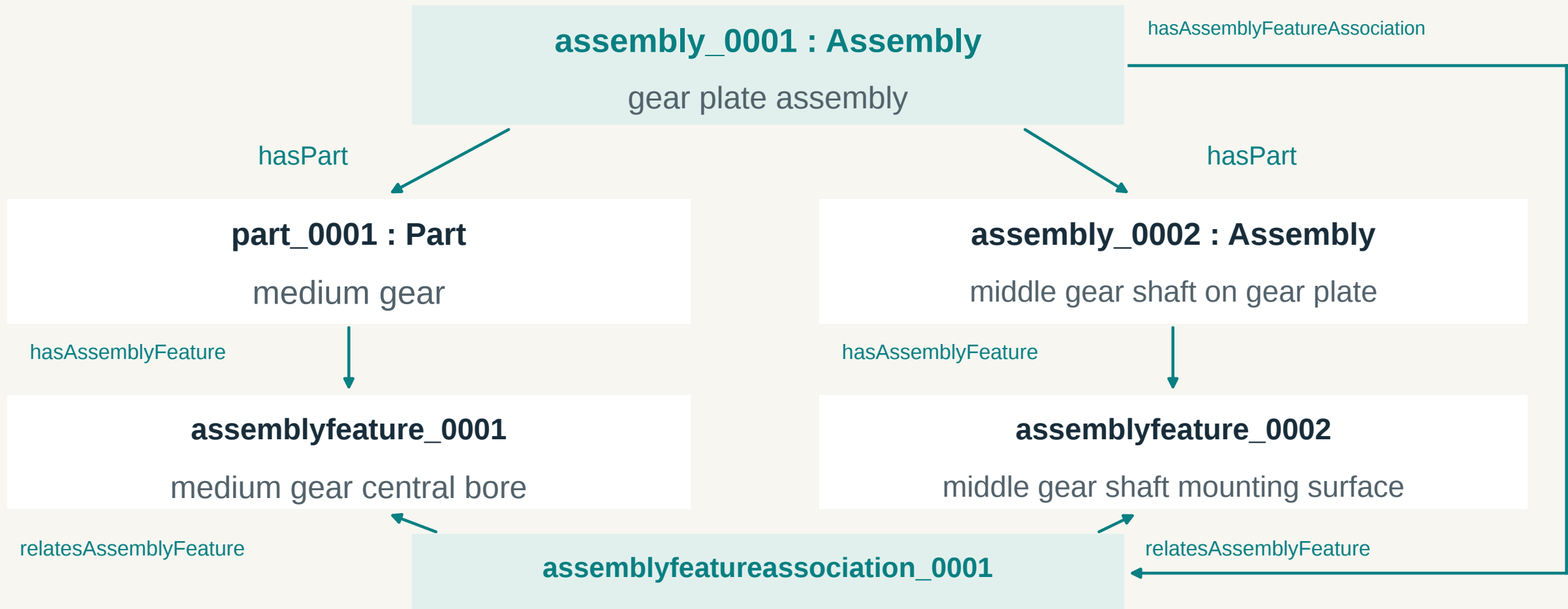
Four properties add ownership and pairwise relations

Domain	Property	Range
Assembly	hasPart	product
product	hasAssemblyFeature	AssemblyFeature
Assembly	hasAssemblyFeatureAssociation	AssemblyFeatureAssociation
AssemblyFeatureAssociation	relatesAssemblyFeature	AssemblyFeature

Each AssemblyFeatureAssociation relates exactly two AssemblyFeature endpoints.

More relationships → more association individuals. Exact owners and endpoints can be shared.

The medium-gear goal becomes concrete instances



Names are proposal annotations; identifiers and edges match the saved ABox. Both endpoint nodes are typed AssemblyFeature.

Relationship state and observation bindings are distinct

Relationship membership

assemblyfeatureassociation_0001 —hasdesiredstate→ desiredstate_0001

Endpoint	state_name	state_value_name
medium gear central bore	current_state	medium gear current location
middle gear shaft mounting surface	desired_state	observed middle gear shaft destination reference

A desired relationship can be anchored by observations of the objects now.

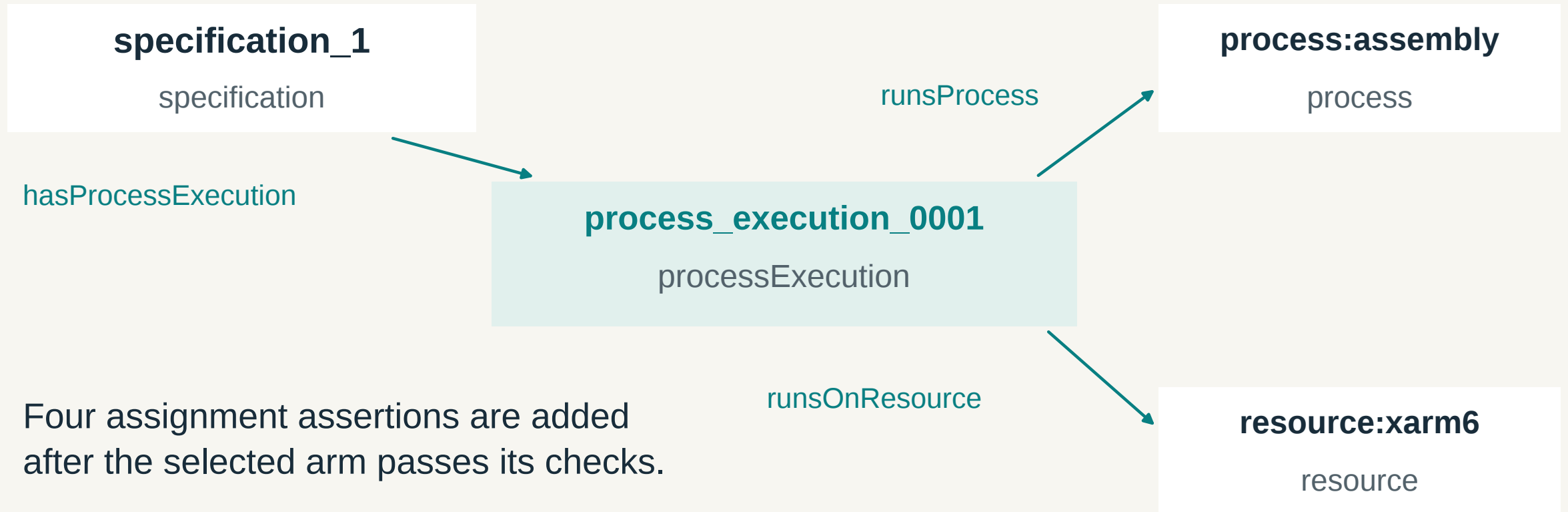
Justify ontology through the contracts it supports

Contract	Explicit representation	Current consumer
Requirement → task	defines + realizes	PA validation / RA handoff
Assembly structure	Ownership + pairwise relations	PA compiler / product context
Resource assignment	capableOf + processExecution	Allocation / selected-RA guard
Evidence accountability	Graph + pinned records	Completion / recovery

Architectural role: implemented. Advantage over equivalent typed JSON: unmeasured.

The ontology describes task structure. PA interprets evidence; MoveIt checks positions; later stages compose and execute.

processExecution records the selected assignment



Four assignment assertions are added after the selected arm passes its checks.

Saved assignment exists; the same completion records motion_executed: false.

OWL semantics and runtime validation are different

Declaration / meaning	What the host explicitly checks
Domain and range	Required property profiles and permitted proposal types
Functional property	Expected state links / complete current assignment
Exactly two related features	Two distinct endpoint records in each association
Missing information can be unknown	Missing required planning evidence blocks completion

Schema semantics + explicit validation + measured evidence have separate responsibilities.

Standards: W3C OWL 2 Primer §2 and W3C SHACL §3.4. Current project checks are implemented in Python.

The ABox is a compact graph of a richer grounded task

Saved ABox — Turtle excerpt

```
ctx:feature_0001
  a ppr:feature ;
  ppr:hascurrentstate
    ctx:currentstate_0001 ;
  ppr:hasdesiredstate
    ctx:desiredstate_0001 .
```

Linked PA proposal + typed record

desired_state.state_values name:

**observed middle gear shaft
destination reference**

value_ref field excerpt:

"field_path": "/cameras/3/candidates/2"

Geometry, uncertainty and provenance remain in linked records with exact references.

The explicit relations support concrete questions

Question	Relation to inspect	Saved / configured answer
What feature is requested?	specification → defines	feature_0001
What process realizes it?	process → realizes → feature	process:assembly
Which resources are candidates?	resource → capableOf → process	xarm6 and ur5e
Which arm is assigned?	hasProcessExecution → runsOnResource	resource:xarm6
Which interfaces are related?	relatesAssemblyFeature	assemblyfeature_0001 assemblyfeature_0002

Inspection patterns over the representation; current execution of arbitrary OWL reasoning or SPARQL is not claimed.

Compare semantic structure and representation separately

Representation	Question it helps test
Prompt-only narrative	How much does explicit structure contribute?
Typed JSON + the same PPR model	What does OWL/RDF add beyond the same contracts?
PPR graph + linked records	How does the current representation compare?

Hold evidence, identities, relationships, model budget and planning conditions fixed.

Measure accuracy, contract failures, repair effort and cost. No comparison has been run here.

Reference: exact PPR property profiles

Property	Domain	Range	Functional
defines	specification	feature	No
realizes	process	feature	No
hascurrentstate	feature	state	Yes
hasdesiredstate	feature	state	Yes
capableOf	resource	process	No
hasProcessExecution	specification	processExecution	No
runsProcess	processExecution	process	Yes
runsOnResource	processExecution	resource	Yes
hasPart	Assembly	product	No
hasAssemblyFeature	product	AssemblyFeature	No
hasAssemblyFeatureAssociation	Assembly	AssemblyFeatureAssociation	No
relatesAssemblyFeature	AssemblyFeatureAssociation	AssemblyFeature	No

AssemblyFeatureAssociation has qualified cardinality 2 on relatesAssemblyFeature → AssemblyFeature.